

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm (Quality Assurance)

University Theory Examination Dec. 2016

PHQA002 QA, GMP, GLP

Date: 14.12.2016, Wednesday

Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 (A) What is quality assurance? Describe the activities of quality assurance department. **05**

(B) Define the following terms: **05**

- (1) Quality Control
- (2) Reference Standard
- (3) Retention Sample
- (4) Recalls
- (5) Specifications

Q.2 Explain the following in relation to pharmaceutical manufacturing facility: **10**

- (1) Plant Layout,
- (2) Environment Control,
- (3) Clean Area,
- (4) Sanitation,
- (5) Cross Contamination,
- (6) Utilities.

OR

Q.2 Write notes on followings: **10**

- (1) Good Warehousing Practices
- (2) Good Distribution Practices

Q.3 Answer the following (Any three) **15**

- (A) What are cGMP guidelines for recruitment of technical staff in pharmaceutical manufacturing area?
- (B) Enumerate GLP guidelines practiced in India and briefly explain scope of GLP activities.
- (C) Define the following terms in relation to GLP: Sponsor, Quality Assurance Unit, Animal care, Protocol for conduct of study.
- (D) Enlist the instruments required for QC laboratory.

- Q.4** (A) Differentiate audit, inspection and periodic evaluation. Give examples. 05
 (B) Explain the benefits of GMP Audit Program. 05
- Q.5** Conceptualize the approaches: (1) Quality by Design, (2) Process Analytical Technology, (3) Lean Six Sigma, (4) Design of Experiment, and (5) Statistical Process Control. 10
- Describe and give conclusion on the relationship between these five approaches.
- OR**
- Q.5** What are the objectives of a stability study? How shelf-life and half-life of pharmaceutical formulation can be predicted from stability data. Discuss the importance of accelerated stability studies. 10
- Q.6** Answer the following (Any three) 15
 (A) Describe QTPPs and CQAs.
 (B) Describe elements of a systematic quality audit program with details.
 (C) Explain the role of QRM in QbD.
 (D) Write a brief note on regulatory aspects of Process Analytical Technology.